

LESSON 4: FLOATING BODIES & METACENTRE

Bodies immersed wholly or partially in a fluid are subjected to upward force which tend to lift (buoy) them upwards.

The *tendency of immersed bodies to be lifted up in the fluid due to an upward force opposite the action of gravity is defined as **buoyancy** and the force doing the lifting is known as the **buoyant force / force of buoyancy / upthrust***

The magnitude of the buoyant force can be determined by ***Archimedes principle*** which states that:

“ When a body is immersed in a fluid either wholly or partially, it is buoyed or lifted up by a force which is equal to the weight of fluid displaced by the body.”

Centre of Buoyancy

Centre of buoyancy – is the point of application of the force of buoyancy on the floating body.

By the following examples, let us appreciate the centre of buoyancy:

Example 1: Consider a wooden block of width 1.25 m, depth 0.75 m, and length 3.0 m floating in water. Taking the specific weight of the wood as 6.4 kN/m^3 . Find:

(i) Volume of water displaced, (ii) Position of centre of buoyancy

Solution. Width of the wooden block = 1.25 m

Depth of the wooden block = 0.75 m

Length of the wooden block = 3.0 m

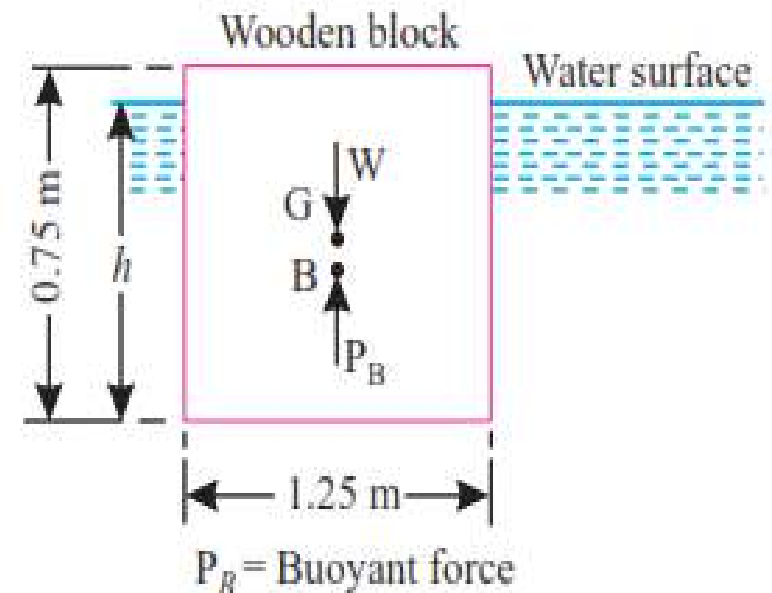
Volume of the block = $1.25 \times 0.75 \times 3 = 2.812 \text{ m}^3$

Specific weight of wood, $w = 6.4 \text{ kN/m}^3$

Weight of the block = $6.4 \times 2.812 = 18 \text{ kN}$

(i) **Volume of water displaced:**

For equilibrium the weight of water displaced
= Weight of wooden block = 18 N



Centre of Buoyancy

Volume of water displaced

$$= \frac{\text{Weight of water displaced}}{\text{Weight density of water}}$$

$$= \frac{18}{9.81} = 1.835 \text{ m}^3 \text{ (Ans)} \quad (\because \text{Weight density of water} = 9.81 \text{ kN/m}^3)$$

(ii) Position of centre of buoyancy:

We know that,

Volume of wooden block in water = Volume of water displaced.

or, $1.25 \times h \times 3.0 = 1.835$

(where, h = depth of wooden block in water)

$$\therefore h = \frac{1.835}{1.25 \times 3.0} = 0.489 \text{ m}$$

$$\text{Hence centre of buoyancy} = \frac{0.489}{2} = 0.244 \text{ from the base (Ans.)}$$

Example 2: Find the density of a metallic body which floats at the surface of mercury of specific gravity 13.6 and water such that 35% of its volume is submerged in mercury and 65% in water.

Solution. Let, V = Volume of the body, m^3 .
Then, volume of body submerged in mercury

$$= \frac{35}{100} \times V = 0.35 V \text{ m}^3$$

Volume of body submerged in water

$$= \frac{65}{100} \times V = 0.65 V \text{ m}^3$$

The body will be in equilibrium when,

Total buoyant (upward) force = weight of the body

$$\begin{aligned} \text{But, Total buoyant force} &= \text{Force of buoyancy due to water} + \text{force of buoyancy due to mercury} \\ &= \text{weight of water displaced by the body} + \text{weight of mercury displaced by the body} \\ &= (\text{weight density of water} \times \text{volume of water displaced}) + (\text{weight density of mercury} \times \text{volume of mercury displaced}) \\ &= 9.81 \times 0.65 V \text{ (kN)} + 13.6 \times 9.81 \times 0.35 V \text{ (kN)} \end{aligned}$$

$$\begin{aligned} \text{and,} \quad \text{Weight of the body} &= \text{weight density} \times \text{volume of the body} \\ &= w_{\text{body}} \times V \end{aligned}$$

(where, w_{body} = weight density of the metallic body)

For equilibrium, we have:

$$9.81 \times 0.65 V + 13.6 \times 9.81 \times 0.35 V = w_{\text{body}} \times V$$

$$\therefore w_{\text{body}} = 53.07 \text{ kN/m}^3 \text{ (Ans.)}$$

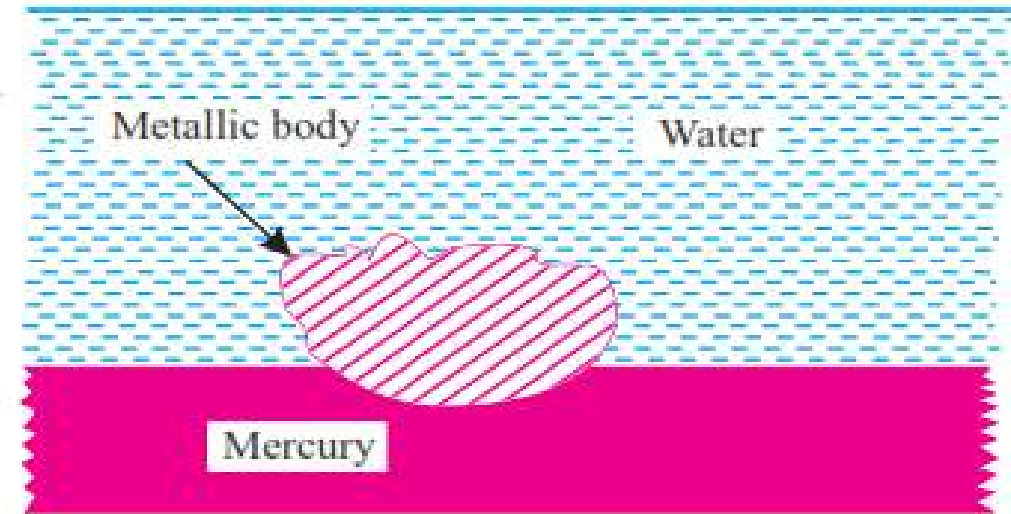


Fig. 4.2

Example 3: A metallic cube 30 cm side and weighing 450 N is lowered into a tank containing a two-fluid layer of water and mercury. Determine the position of block at mercury-water interface when it has reached equilibrium.

Solution. Refer to Fig. 4.3. The metallic cube sinks beneath the water surface and comes to rest at the water-mercury interface.

As per principle of floatation, we have weight of cubical block = buoyant force

= weight of water and mercury displaced by the block.

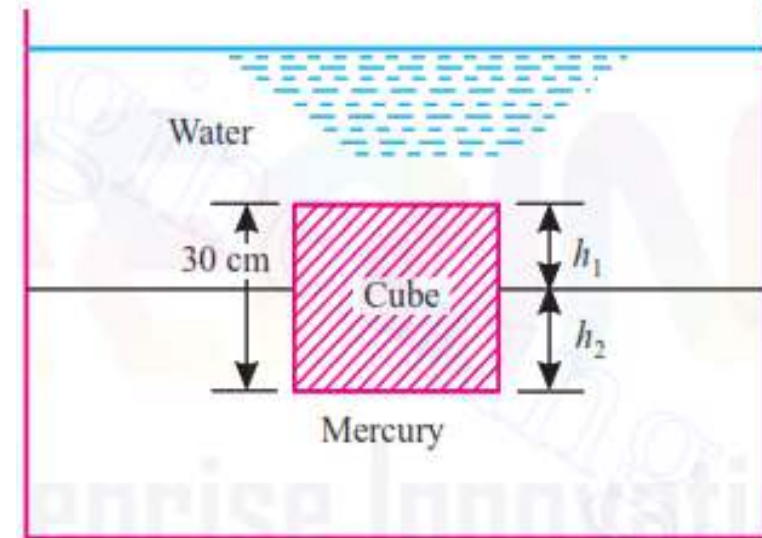


Fig. 4.3

$$\begin{aligned}\text{Thus,} \quad 450 &= 9810 (h_1 \times 0.3 \times 0.3) + 9810 \times 13.6 (h_2 \times 0.3 \times 0.3) \\ &= (h_1 + 13.6 h_2) \times (9810 \times 0.3 \times 0.3)\end{aligned}$$

$$\text{or,} \quad (h_1 + 13.6 h_2) = \frac{450}{9810 \times 0.3 \times 0.3} = 0.509 \text{ m} \quad \dots(i)$$

$$\text{Also,} \quad h_1 + h_2 = 0.3 \text{ m} \quad \dots(ii)$$

From (i) and (ii), we have the depth of cube below the water-mercury interface,

$$h_2 = \frac{(0.509 - 0.3)}{12.6} = 0.01658 \text{ m or } 16.58 \text{ mm (Ans.)}$$

Example 4: A cylinder of mass 10 kg and area of cross-section 0.1 m^2 is tied down with string in a vessel containing two liquids as shown in figure 4.7. Calculate gauge pressure on the cylinder bottom and the tension in the string. Density of water = 1000 kg/m^3 . Specific gravity of A=0.8. B (water) = 1.0

Solution. Given: Mass of cylinder, $m = 10 \text{ kg}$

Area of cross-section = 0.1 m^2

Density of water (liquid B) = 1000 kg/m^3

Density of liquid A = $0.8 \times 1000 = 800 \text{ kg/m}^3$

Tension in string, T :

$$\begin{aligned}\text{Volume of liquid A displaced} &= 0.1 \times 0.1 \\ &= 0.01 \text{ m}^3\end{aligned}$$

$$\therefore \text{Mass of liquid A displaced, } m_A = 0.01 \times 800 = 8 \text{ kg}$$

$$\therefore \text{Volume of liquid B displaced} = 0.1 \times 0.125 = 0.0125 \text{ m}^3$$

$$\therefore \text{Mass of liquid B displaced, } m_B = 0.0125 \times 1000 = 12.5 \text{ kg}$$

$$\text{Total mass of liquid displaced} = m_A + m_B = 8 + 12.5 = 20.5 \text{ kg}$$

$$\text{Upward thrust} = 20.5 \times 9.81 = 201.1 \text{ N}$$

$$\text{Weight of cylinder} = mg = 10 \times 9.81 = 98.1 \text{ N}$$

$$\text{Net upward thrust} = 201.1 - 98.1 = 103 \text{ N}$$

$$\therefore \text{Tension in the string, } T = 103 \text{ N (Ans.)}$$

Pressure (gauge) on the cylinder bottom, p :

$$p = \frac{\text{Net upward thrust}}{\text{Area of cross-section}} = \frac{103}{0.1} = 1030 \text{ N/m}^2 \text{ (Ans.)}$$

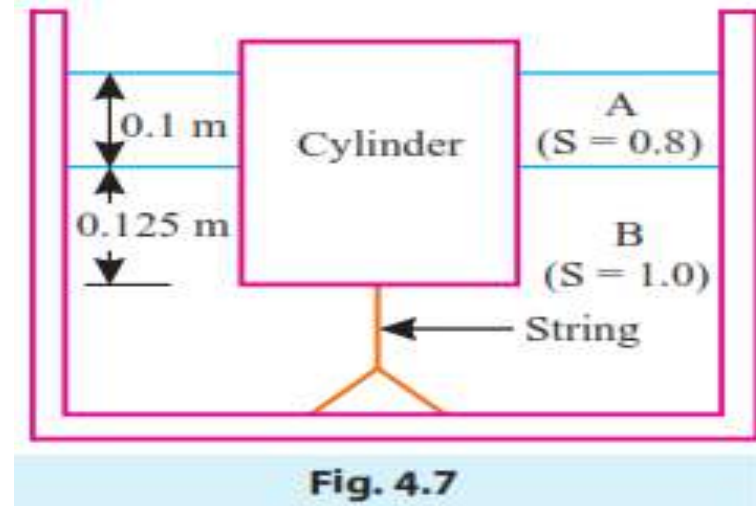


Fig. 4.7

Types of Equilibrium for Floating Bodies in Fluids

Stable Equilibrium– When a body is given a small angular displacement (tilted slightly), by some external force, and then it returns back to its original position due to the internal forces (the weight and the upthrust),.

Unstable Equilibrium– If a body does not return to its original position from the slightly displaced angular position and heels farther away, when given a small angular displacement.

Neutral Equilibrium– If a body, when given a small angular displacement, occupies a new position and remains at rest in that new position.

Meta-centre and Metacentric Height

Meta-Centre— This is the point about which a body starts to oscillate when tilted by a small angle. It may also be defined as the point at which the line of action of the force of buoyancy meets the normal axis of the body when the body is given a small angular displacement.

Consider a body floating in a fluid as shown in fig 4.5 (a) with G being the centre of gravity and B the centre of buoyancy. For equilibrium, the two points lie on the normal vertical axis

Under a small angular displacement in the clockwise direction as shown in (b), The centre of buoyancy shifts away from the c.o.g. The line of action of force of buoyancy in the new position will intersect the normal axis of the body at some point say M . This point M is called metacentre

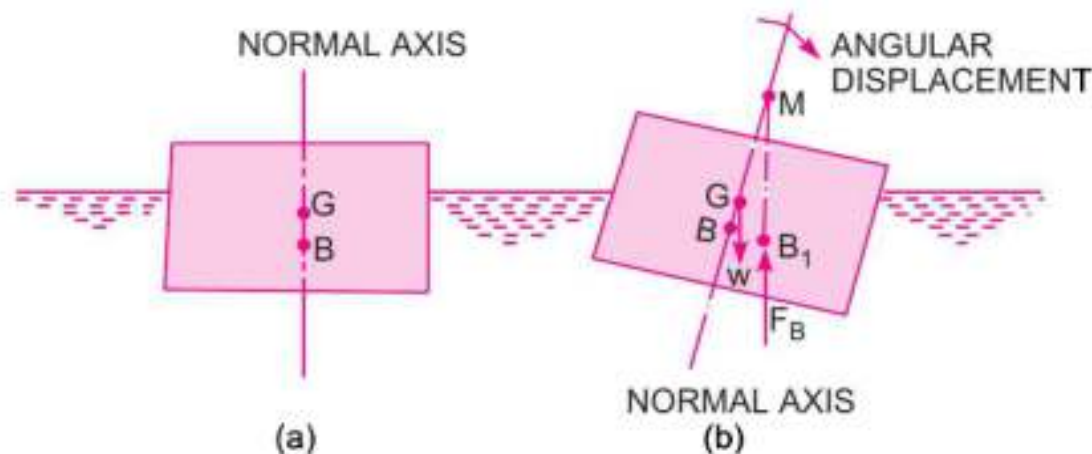


Fig. 4.5 *Meta-centre*

Meta-centre and Metacentric Height

Metacentric Height – The distance between the centre of gravity of a floating body and the metacentre (i.e. distance GM as shown in Fig.4.5 (b) is called metacentric height.

- For stable equilibrium, the position of metacentre M remains higher than c.g. of the body, G .
- For unstable equilibrium, the position of metacentre M remains lower than G.
- For neutral equilibrium, the position of metacentre M coincides with G

Determination of Metacentric Height

Metacentric height may be determined by the following two methods:

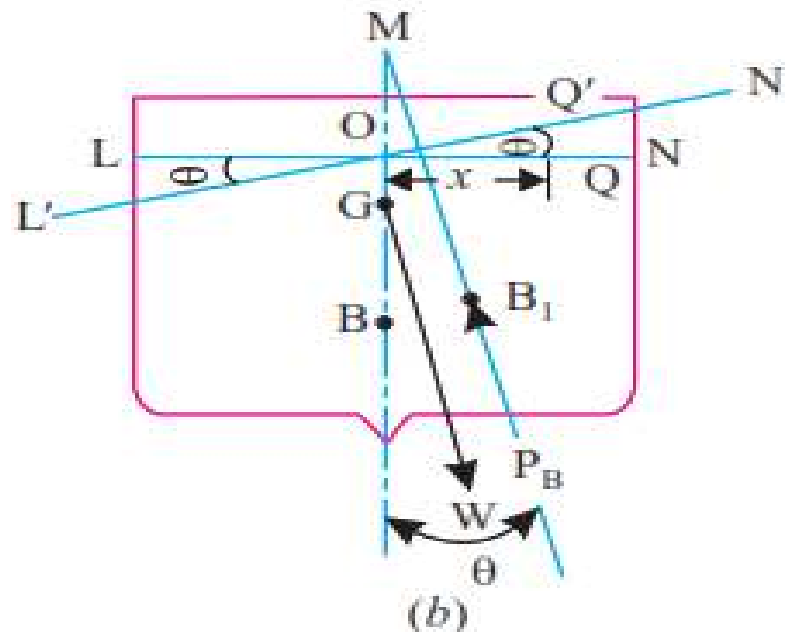
1. Analytical method
2. Experimental method

1. Analytical Method

Refer to Fig. 4.6 (b) with line $L'ON$ representing the water surface. Position $L'OL$ of the body is submerged and the position $N'ON$ is lifted due to tilting. As a result of this, the centre of buoyancy changes its position from B to B_1 . The intersection of axis of the body and the vertical line through B_1 locates the metacentre, M of the body.

To determine the metacentric height GM , consider an elementary cylindrical prism QQ' of portion $N'ON$ at a distance " x " from O .

Let the area of this elementary prism be dA . The height is given by $x.\theta$. The volume of the elementary prism is given by:



$$\delta V = x.\theta.\delta A. \quad \dots(i)$$

The upward force or buoyancy force acting at this prism (δP_B) is given by:

$$\delta P_B = w.\delta V = w. x.\theta.\delta A \quad \dots(ii)$$

(where, w = unit weight of liquid)

The moment of this buoyancy force about O

$$x.\delta P = w.\theta.x^2.\delta A \quad \dots(iii)$$

For the total portion $N'ON$, this moment is given by:

$$\int x. dP_B = \int w.\theta. x^2. dA = w\theta \int x^2 dA \quad \dots(iv)$$

or,
$$\int x. dP_B = w\theta.I$$

(where, I = moment of inertia of the sectional area at the water line about the axis through O)

$\int x. dP_B$ gives the change in moment due to buoyancy.

Now,
$$\int x. dP_B (= w.\theta.I) = P_B \times BB_1$$

(where, P_B = total force of buoyancy)

But,
$$BB_1 = BM \times \theta \text{ and } P_B = W = w \times V$$

$$\therefore w.\theta.I = w.V.BM.\theta \quad \text{or} \quad BM = \frac{I}{V} \quad \dots(4.1)$$

Now metacentric height, $GM = BM \pm BG$

+ ve sign : when G is *lower* than B

– ve sign : when G is *higher* than B

1. Experimental Method

Refer to Fig. 4.9 below. In this method, a known weight W_1 is shifted by distance z across the axis of tilt. The change in moment due to this shift is $W_1 z$. Let the tilt be θ . This angle of tilt may be measured experimentally by using a plumb bob. The change in momentum due to this tilt is equal to $W.GM \tan \theta$

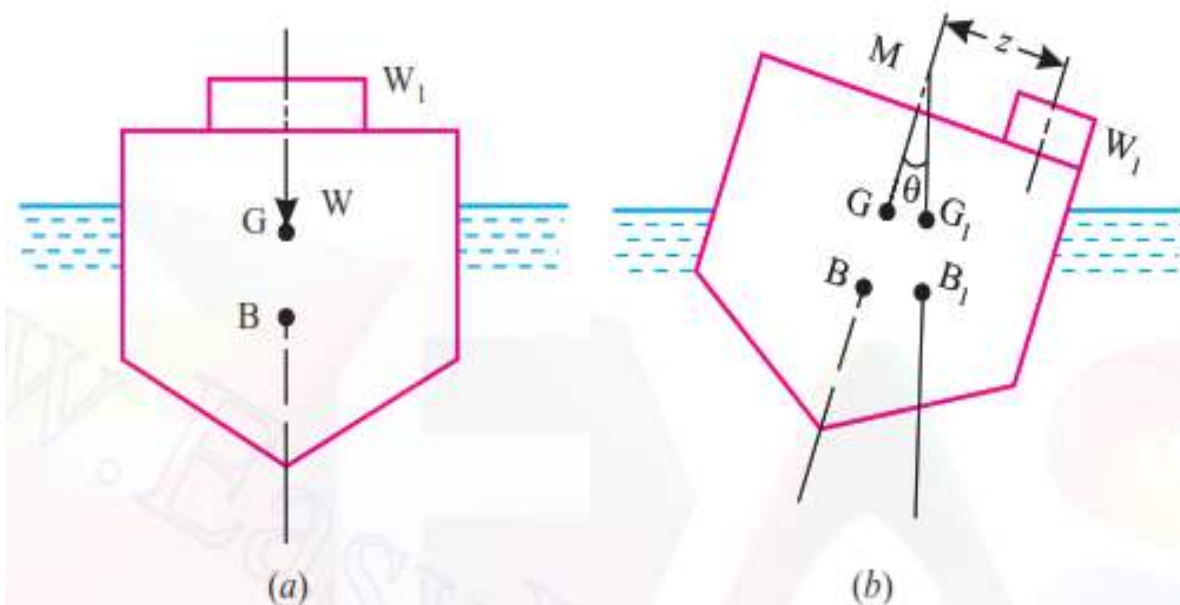


Fig. 4.9. Experimental method for determination of metacentric height.

$$W_1 \cdot z = W \cdot GM \cdot \tan \theta \quad \text{or} \quad GM = \frac{W_1 \cdot z}{W \cdot \tan \theta}$$

l = Length of plumb bob, and

d = Displacement of the plumb bob,

Then, $\tan \theta = \frac{d}{l}$

and, metacentric height is given by:

$$GM = \frac{W_1 \cdot z \cdot l}{W \cdot d} \quad \dots(4.3)$$

Example 1: A wooden block of specific gravity 0.75 floats in water. If the size of the block is 1 m x 0.5 m x 0.4 m, find its metacentric height.

Solution. Size (or dimensions) of the block = 1 m
 × 0.5 m × 0.4 m

Specific gravity of wood = 0.75

Specific weight $w = 0.75 \times 9.81 = 7.36 \text{ kN/m}^3$

Weight of wooden block = specific weight × volume
 = $7.36 \times 1 \times 0.5 \times 0.4 = 1.472 \text{ kN}$

Let depth of immersion = h metres.

Weight of water displaced

= Specific weight of water × volume of the wood submerged in water

$$= 9.81 \times 1 \times 0.5 \times h = 4.9 h \text{ kN}$$

Now, for equilibrium:

Weight of wooden block = Weight of water displaced i.e., $1.472 = 4.9 h$

or,
$$h = \frac{1.472}{4.9} = 0.3 \text{ m}$$

∴ Distance of centre of buoyancy from bottom i.e.,

$$OB = \frac{h}{2} = \frac{0.3}{2} = 0.15 \text{ m}$$

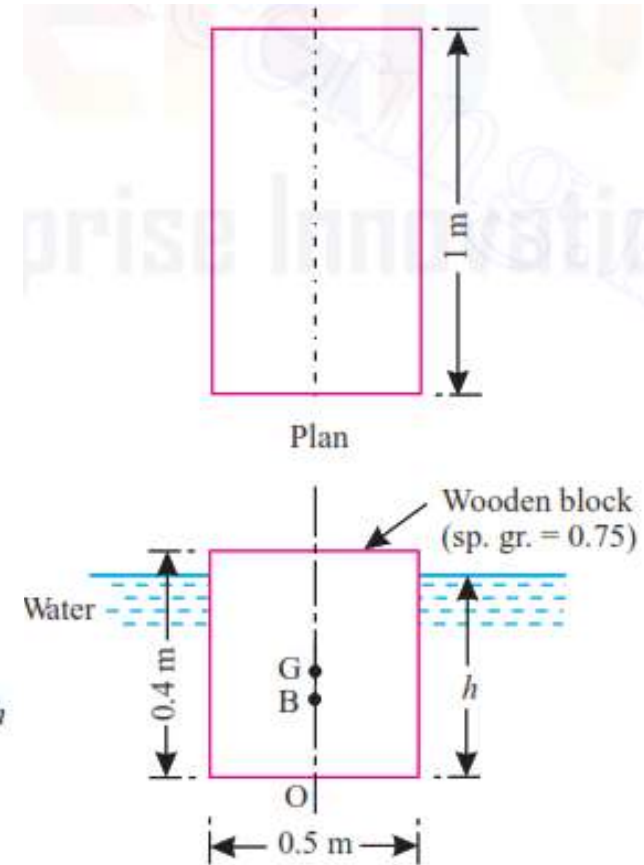
and,
$$OG = \frac{0.4}{2} = 0.2 \text{ m}$$

∴
$$BG = OG - OB = 0.2 - 0.15 = 0.05 \text{ m}$$

Also,
$$BM = \frac{I}{V}$$

Where, I = Moment of inertia of a rectangular section

$$= \frac{1 \times 0.5^3}{12} = 0.014 \text{ m}^4$$



and, V = Volume of water displaced (or volume of wood in water)

$$= 1 \times 0.5 \times h = 1 \times 0.5 \times 0.3 = 0.15 \text{ m}^3$$

$$BM = \frac{I}{V} = \frac{0.014}{0.15} = 0.093 \text{ m}$$

We know that the metacentric height,

$$GM = BM - BG$$

$$= 0.093 - 0.05 = 0.043 \text{ m (Ans.)}$$

Example 2: A solid cube of sides 0.5 m each is made of a material of relative density 0.5. The cube floats in a liquid of relative density of 0.95 with two of its faces horizontal. Examine its stability.

Depth of cube in liquid,

$$h = \frac{0.5 \times 0.5}{0.95} = 0.263 \text{ m}$$

Distance of centre of buoyancy (B) from O ,

$$OB = \frac{0.263}{2} = 0.1315 \text{ m}$$

Distance of centre of gravity (G) from O ,

$$OG = \frac{0.5}{2} = 0.25 \text{ m}$$

$$BG = OG - OB = 0.25 - 0.1315 \\ = 0.1185 \text{ m}$$

B lies below G .

$$BM = \frac{I}{V}$$

where, I = Moment of inertia of the plane of the body about YY

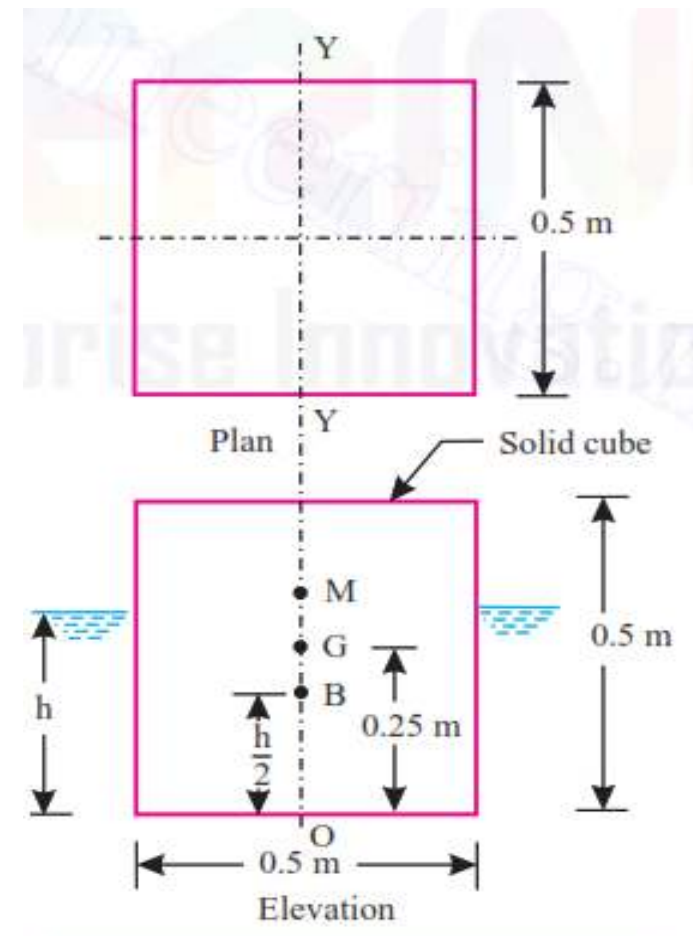
$$= \frac{1}{12} (0.5) (0.5)^3 = 0.005208 \text{ m}^4$$

V = Volume of liquid displaced

$$= 0.5 \times 0.5 \times 0.263 = 0.06575 \text{ m}^3$$

$$BM = \frac{I}{V} = \frac{0.005208}{0.06575} = 0.0792 \text{ m}$$

Metacentric height, $GM = BM - BG = 0.0792 - 0.1185 = -0.0393 \text{ m}$



Example 3: A solid cylinder of diameter 4.0 m has a height of 3 m. Find the metacentric height of the cylinder when it is floating in water with its axis vertical. The sp gr of the cylinder = 0.6.

Solution. Given :

Dia. of cylinder, $D = 4.0 \text{ m}$

Height of cylinder, $h = 3.0 \text{ m}$

Sp. gr. of cylinder = 0.6

Depth of immersion of cylinder
 $= 0.6 \times 3.0 = 1.8 \text{ m}$

$$\therefore AB = \frac{1.8}{2} = 0.9 \text{ m}$$

and $AG = \frac{3}{2} = 1.5 \text{ m}$

$$\therefore BG = AG - AB$$

$$= 1.5 - 0.9 = 0.6 \text{ m}$$

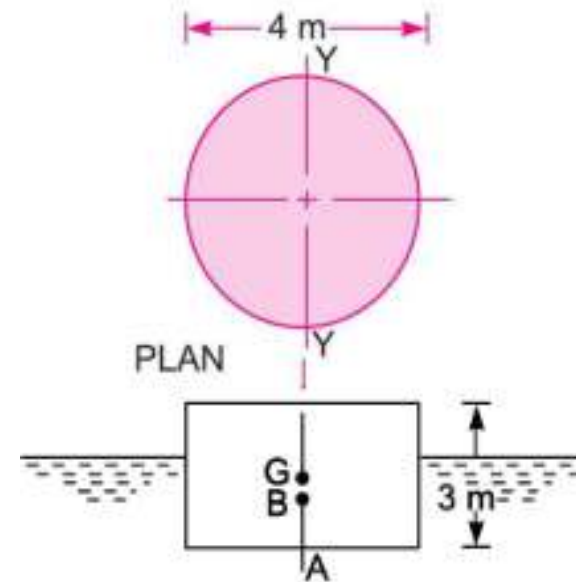
Now the meta-centric height GM is given by equation

$$GM = \frac{I}{\nabla} - BG$$

But

$I = \text{M.O.I. about } Y-Y \text{ axis of the plan of the body}$

$$= \frac{\pi}{64} D^4 = \frac{\pi}{64} \times (4.0)^4$$



and

∇ = Volume of cylinder in water

$$= \frac{\pi}{4} D^2 \times \text{Depth of immersion}$$

$$= \frac{\pi}{4} (4)^2 \times 1.8 \text{ m}^3$$

$\therefore$

$$GM = \frac{\frac{\pi}{64} \times (4.0)^4}{\frac{\pi}{4} \times (4.0)^2 \times 1.8} - 0.6$$

$$= \frac{1}{16} \times \frac{4.0^2}{1.8} - 0.6 = \frac{1}{1.8} - 0.6 = 0.55 - 0.6 = -0.05 \text{ m. Ans.}$$

– ve sign means that meta-centre, (M) is below the centre of gravity (G).

Example 4: A block of wood of specific gravity 0.7 floats in water. Determine the metacentric height of the block if its size is 2 m x 1 m x 0.8 m.

Solution. Given :

Dimension of block = 2 × 1 × 0.8

Let depth of immersion = h m

Sp. gr. of wood = 0.7

Weight of wooden piece = Weight density of wood × Volume
 = $0.7 \times 1000 \times 9.81 \times 2 \times 1 \times 0.8$ N

Weight of water displaced = Weight density of water
 × Volume of the wood sub-merged in water
 = $1000 \times 9.81 \times 2 \times 1 \times h$ N

For equilibrium,

Weight of wooden piece = Weight of water displaced

$$\therefore 700 \times 9.81 \times 2 \times 1 \times 0.8 = 1000 \times 9.81 \times 2 \times 1 \times h$$

$$\therefore h = \frac{700 \times 9.81 \times 2 \times 1 \times 0.8}{1000 \times 9.81 \times 2 \times 1} = 0.7 \times 0.8 = 0.56 \text{ m}$$

$\therefore$ Distance of centre of Buoyancy from bottom, i.e.,

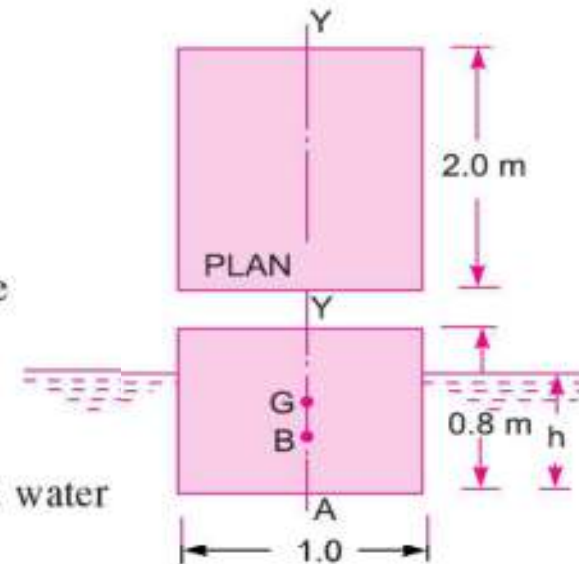
$$AB = \frac{h}{2} = \frac{0.56}{2} = 0.28 \text{ m}$$

and $AG = 0.8/2.0 = 0.4$ m

$$\therefore BG = AG - AB = 0.4 - 0.28 = 0.12 \text{ m}$$

The meta-centric height is given by equation (4.4) or

$$GM = \frac{I}{\nabla} - BG$$



where $I = \frac{1}{12} \times 2 \times 1.0^3 = \frac{1}{6} \text{ m}^4$

$$\nabla = \text{Volume of wood in water} \\ = 2 \times 1 \times h = 2 \times 1 \times .56 = 1.12 \text{ m}^3$$

$$GM = \frac{1}{6} \times \frac{1}{1.12} - 0.12 = 0.1488 - 0.12$$

$$= 0.0288 \text{ m. Ans.}$$

HIGHLIGHTS

1. The upward force exerted by a liquid on a body when the body is immersed in the liquid is known as buoyancy or force of buoyancy.
2. The point through which force of buoyancy is supposed to act is called centre of buoyancy.
3. The point about which a body starts oscillating when the body is tilted is known as meta-centre.
4. The distance between the meta-centre and centre of gravity is known as meta-centric height.

5. The meta-centric height (GM) is given by $GM = \frac{I}{\nabla} - BG$

where I = Moment of Inertia of the floating body (in plan) at water surface about the axis $Y-Y$

∇ = Volume of the body sub-merged in water

BG = Distance between centre of gravity and centre of buoyancy.

6. Conditions of equilibrium of a floating and sub-merged body are :

<i>Equilibrium</i>	<i>Floating Body</i>	<i>Sub-merged Body</i>
(i) Stable Equilibrium	M is above G	B is above G
(ii) Unstable Equilibrium	M is below G	B is below G
(iii) Neutral Equilibrium	M and G coincide	B and G coincide

7. The value of meta-centric height GM , experimentally is given as $GM = \frac{w_1 x}{W \tan \theta}$

where w_1 = Movable weight

x = Distance through which w_1 is moved

W = Weight of the ship or floating body including w_1

θ = Angle through the ship or floating body is tilted due to the movement of w_1 .

8. The time period of oscillation or rolling of a floating body is given by $T = 2\pi \sqrt{\frac{K^2}{GM \times g}}$

where K = Radius of gyration, GM = Meta-centric height

T = Time of one complete oscillation.